Exploratory Data Analysis (EDA)

Pada tahap ini dilakukan eksplorasi distribusi data untuk memahami karakteristik masing-masing fitur serta mendeteksi keberadaan outlier menggunakan histogram dan boxplot.

```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
import seaborn as sns
from sklearn.model_selection import train_test_split

# Scalling untuk MinMax Scaller
from sklearn.preprocessing import MinMaxScaler
# Set the maximum number of columns and rows to display to a large number
pd.set_option('display.max_columns', None)
pd.set_option('display.max_rows', None)
```

B. Outlier Handling

Outlier Handling : InterQuartile Handling

```
df_california = pd.read_csv('/content/sample_data/california_housing_train.csv')
```

```
# untuk plot Q-Q
import scipy.stats as stats
```

```
df_california.head()
```

	longitude	latitude	housing_median_age	total_rooms	total_bedrooms	population	households	median_income	median_house_
0	-114.31	34.19	15.0	5612.0	1283.0	1015.0	472.0	1.4936	6
1	-114.47	34.40	19.0	7650.0	1901.0	1129.0	463.0	1.8200	8
2	-114.56	33.69	17.0	720.0	174.0	333.0	117.0	1.6509	8
3	-114.57	33.64	14.0	1501.0	337.0	515.0	226.0	3.1917	7
4	-114.57	33.57	20.0	1454.0	326.0	624.0	262.0	1.9250	6

Next steps: Generate code with df_california New interactive sheet

```
from sklearn.model_selection import train_test_split
# splitting terlebih dahulu


#NOTES :
#train : test = 80:20 atau 75:25 atau (minimum :70:30 atau maksimum: 90:10) bagi angka yg lain (85:15)
#data train di-handling outliernya
#tapi tidak untuk data test, karena data test ibarat representasi data masa depan yang digunakan untuk melihat performa Ma
```

```
df_california.columns.tolist()
```

```
['longitude',
 'latitude',
 'housing_median_age',
 'total_rooms',
 'total_bedrooms',
 'population',
 'households',
 'median_income',
 'median_house_value']
```

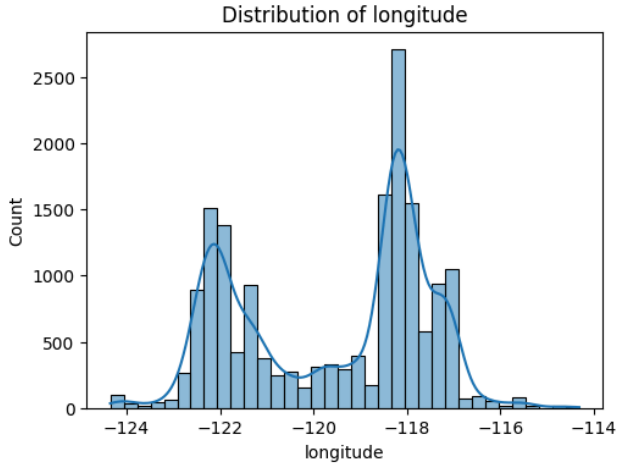
Assignment 1.a :

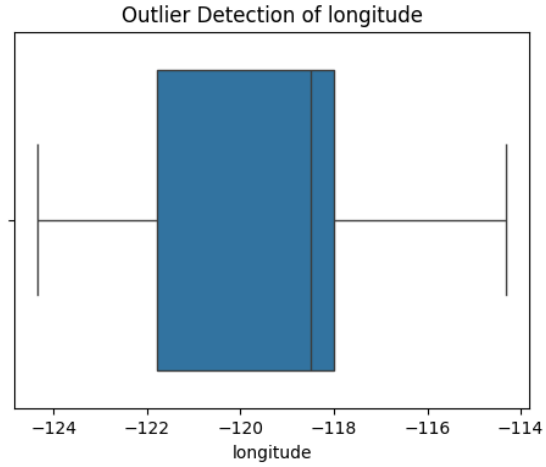
1. lakukan looping untuk melihat plot distribusi data dan outlier pada kolom-kolom : ['MedInc', 'HouseAge', 'AveRooms','AveBedrms', 'AveOccup'] di california_dataset.csv
2. Daftar nama kolom yang Anda cek, kolom_kolom = ['MedInc', 'HouseAge', 'AveRooms', 'AveBedrms', 'AveOccup']

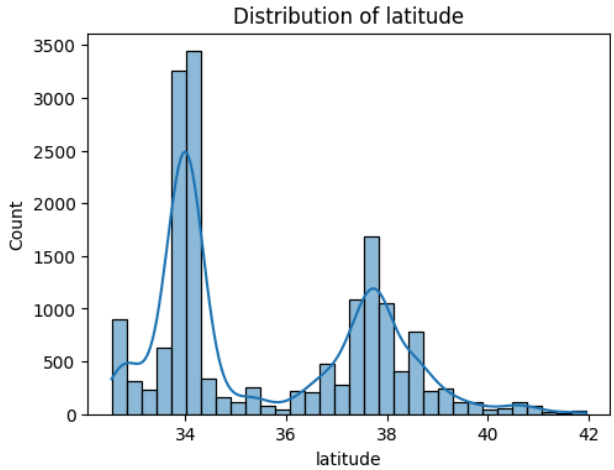
```
df_california.head()
df_california.columns
```

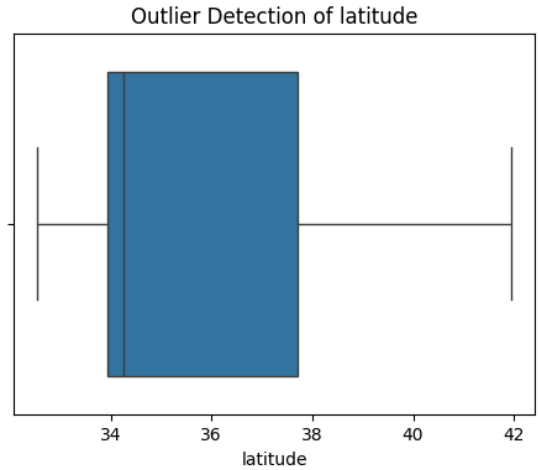
```
Index(['longitude', 'latitude', 'housing_median_age', 'total_rooms',
       'total_bedrooms', 'population', 'households', 'median_income',
       'median_house_value'],
      dtype='object')
```

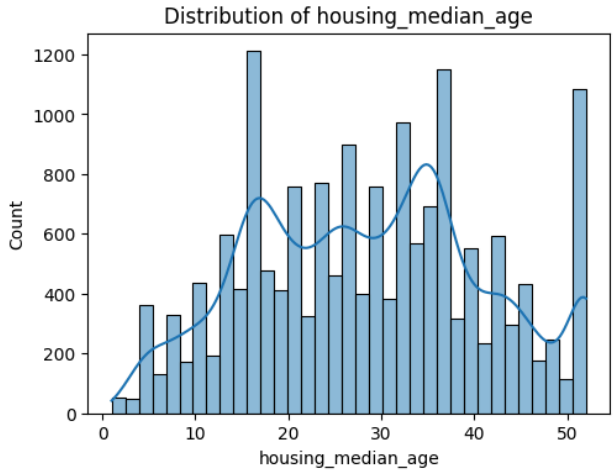
```
columns = df_california.columns

for col in columns:

    if df_california[col].dtype != 'object':

        plt.figure(figsize=(12,4))

        plt.subplot(1,2,1)
        sns.histplot(df_california[col], kde=True)
        plt.title(f'Distribution of {col}')

        plt.subplot(1,2,2)
        sns.boxplot(x=df_california[col])
        plt.title(f'Outlier Detection of {col}')

        plt.show()
```

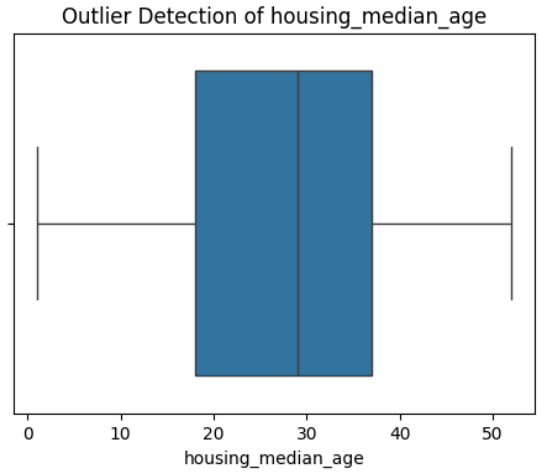

Distribution of longitude
Count
longitude

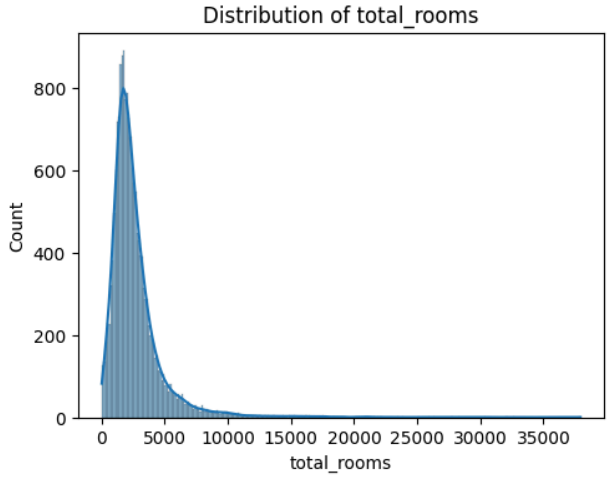

Outlier Detection of longitude
longitude

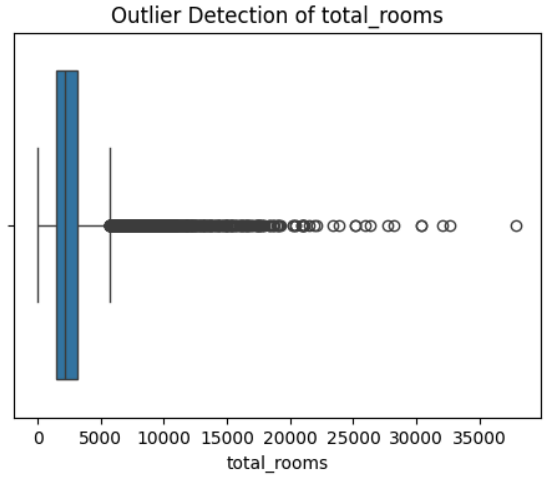

Distribution of latitude
Count
latitude

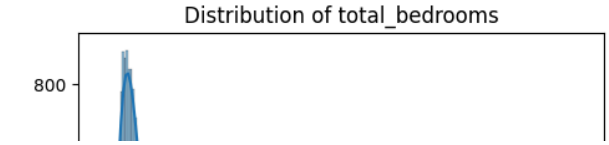

Outlier Detection of latitude
latitude

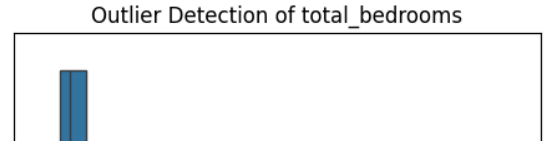

Distribution of housing_median_age
Count
housing_median_age

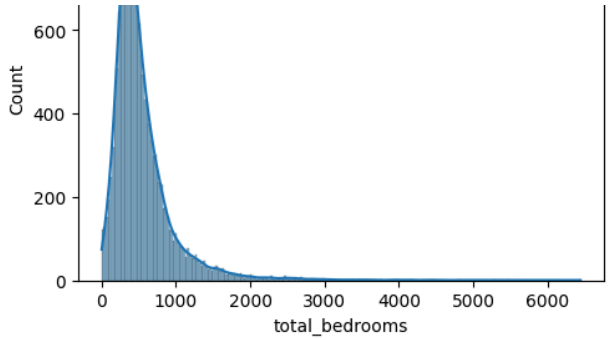

Outlier Detection of housing_median_age
housing_median_age

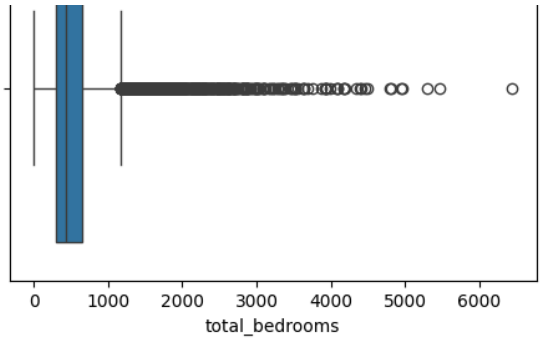

Distribution of total_rooms
Count
total_rooms


Outlier Detection of total_rooms
total_rooms

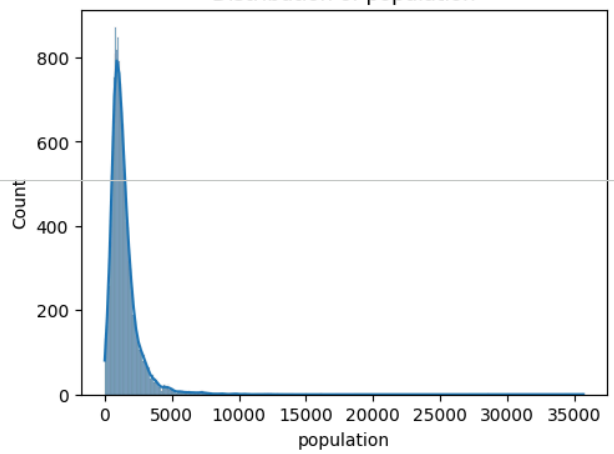

Distribution of total_bedrooms


Outlier Detection of total_bedrooms

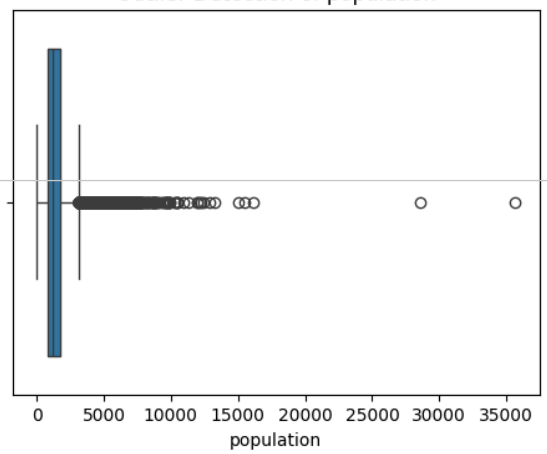




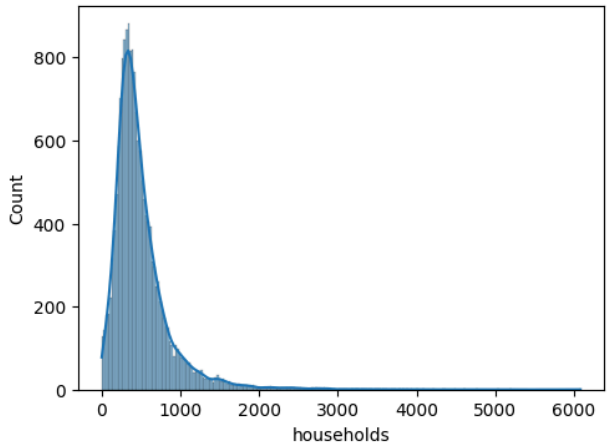
Distribution of population



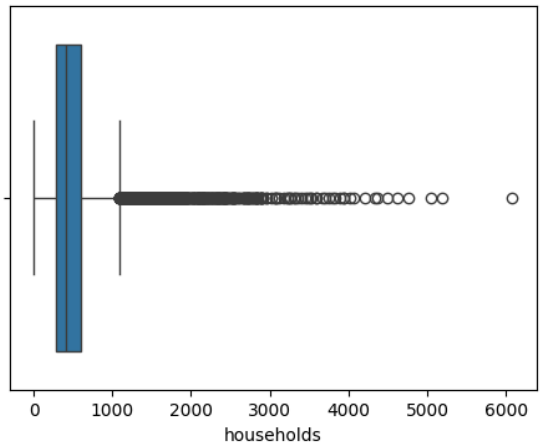
Outlier Detection of population



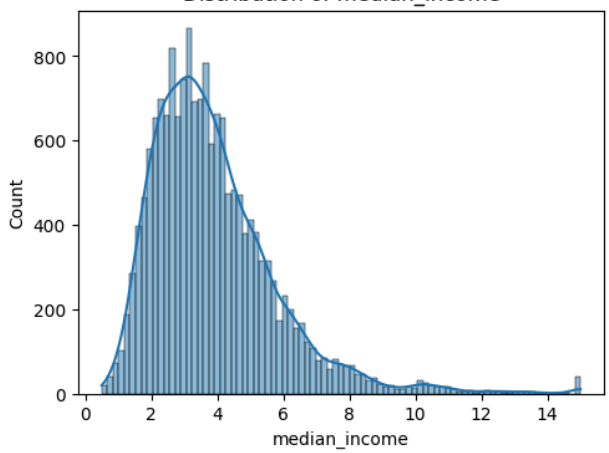
Distribution of households



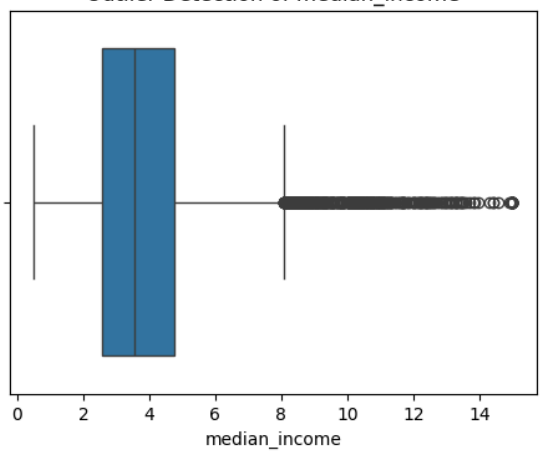
Outlier Detection of households



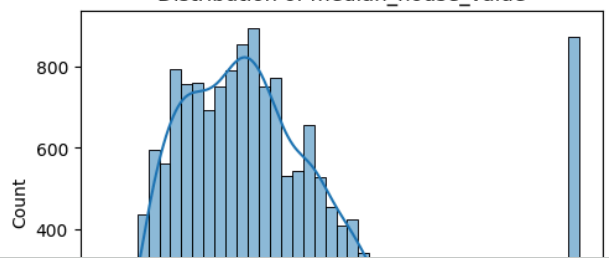
Distribution of median_income



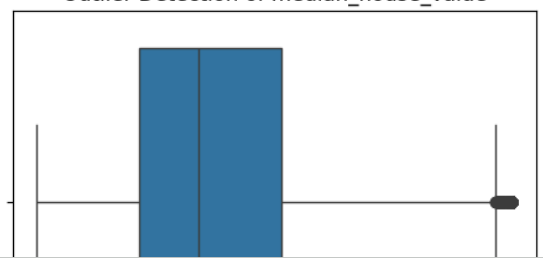
Outlier Detection of median_income



Distribution of median_house_value



Outlier Detection of median_house_value



```
df_california = df_california.rename(columns={'median_income': 'MedInc', 'housing_median_age': 'HouseAge'})
```

```
# Create derived columns AveRooms, AveBedrms, and AveOccup
df_california['AveRooms'] = df_california['total_rooms'] / df_california['households']
df_california['AveBedrms'] = df_california['total_bedrooms'] / df_california['households']
df_california['AveOccup'] = df_california['population'] / df_california['households']

# Display the first few rows to confirm the new column names and derived features
display(df_california.head())
```

	longitude	latitude	HouseAge	total_rooms	total_bedrooms	population	households	MedInc	median_house_value	AveRooms
0	-114.31	34.19	15.0	5612.0	1283.0	1015.0	472.0	1.4936	66900.0	11.889831
1	-114.47	34.40	19.0	7650.0	1901.0	1129.0	463.0	1.8200	80100.0	16.522678
2	-114.56	33.69	17.0	720.0	174.0	333.0	117.0	1.6509	85700.0	6.153846
3	-114.57	33.64	14.0	1501.0	337.0	515.0	226.0	3.1917	73400.0	6.641593
4	-114.57	33.57	20.0	1454.0	326.0	624.0	262.0	1.9250	65500.0	5.549618

Assignment 1.b

1. Lakukan handling outlier pilih 1 kolom saja yang di handling dengan IQR setelah mengecek distribusinya
2. Dan cek distribusinya kembali

```
col = 'total_rooms'
```

```
Q1 = df_california[col].quantile(0.25)
Q3 = df_california[col].quantile(0.75)

print("Q1 :", Q1)
print("Q3 :", Q3)
```

```
Q1 : 1462.0
Q3 : 3151.25
```

```
IQR = Q3 - Q1
print("IQR :", IQR)
```

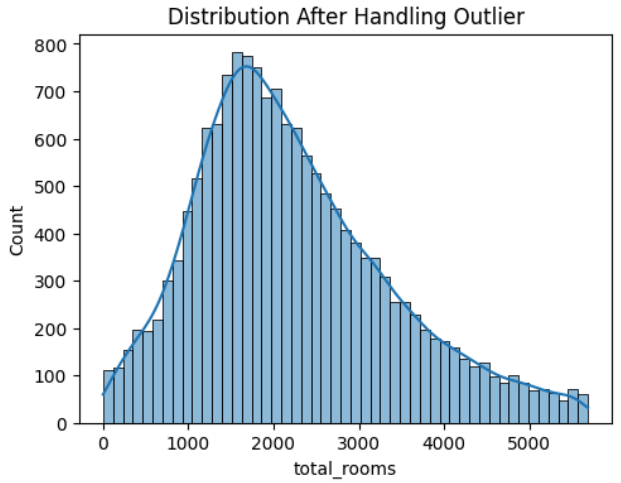
```
IQR : 1689.25
```

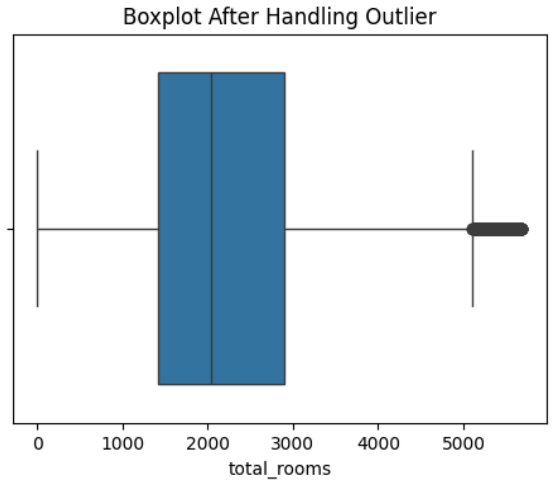
```
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

print("Lower Bound :", lower_bound)
print("Upper Bound :", upper_bound)
```

```
Lower Bound : -1071.875
Upper Bound : 5685.125
```

```
df_california_clean = df_california[
    (df_california[col] >= lower_bound) &
    (df_california[col] <= upper_bound)
]
```

```
plt.figure(figsize=(12,4))

plt.subplot(1,2,1)
sns.histplot(df_california_clean[col], kde=True)
plt.title("Distribution After Handling Outlier")

plt.subplot(1,2,2)
sns.boxplot(x=df_california_clean[col])
plt.title("Boxplot After Handling Outlier")

plt.show()
```





C. Missing Value Handling

Missing Value Merupakan nilai yang hilang di suatu kolom, namun tidak hanya sebatas itu. Untuk nilai yang tidak sesuai tapi muncul didalam suatu kolom tertentu juga bisa dianggap sebagai missing value. Misal tipenya kategorikal tapi isinya ada yang berupa angka dan juga sebaliknya.

Assignment 2. Missing Value Handling :

1. Cek persentase missing value pada kolom Headquarters di company.csv, Cek jenis missing valuenya ?
2. apakah di drop atau tidak (gunakan syarat drop kolom jika > 20%)
3. jika tidak maka handling, dan dengan cara apa handling nilainya, median atau modus ?

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df_company = pd.read_csv('company.csv')
```

```
import os
os.listdir()
```

```
['.config',
 'TelcoCustomerChurn.csv',
 'company.csv',
 'TelcoCustomerChurn.xlsx',
 'sample_data']
```

```
import pandas as pd

df_company = pd.read_csv('company.csv')
```

```
df_company.head()
```

	Revenue	Size	Rating	Headquarters
0	Unknown / Non-Applicable	1001 to 5000 employees	3.1	New York, NY
1	$1 to $2 billion (USD)	5001 to 10000 employees	4.2	Herndon, VA
2	$100 to $500 million (USD)	1001 to 5000 employees	3.8	Boston, MA
3	$100 to $500 million (USD)	501 to 1000 employees	3.5	Bad Ragaz, Switzerland
4	Unknown / Non-Applicable	51 to 200 employees	2.9	New York, NY

Next steps: Generate code with df_company New interactive sheet

```
# cek jumlah missing value pada kolom Headquarters
missing_value = df_company['Headquarters'].isnull().sum()

total_data = len(df_company)
```